The Psychological Sense of School Membership Scale and the Belonging Scale

Description and purpose

These scales assess pupils' sense of belonging (psychological membership) in their school – the extent to which they feel accepted, included, respected and supported. At a whole-school level 'belongingness' or 'school membership' can provide an indicator of the quality of school social relations. At the level of the group or individual it has been identified as a potentially critical factor in promoting the participation in school of 'at-risk' pupils. Where pupils do not feel a sense of membership in their school as a social system their motivation, engagement, academic achievement and attendance are at all at risk (Goodenow 1993).

The *Psychological Sense of School Membership Scale* (*PSSM*) was developed for pupils aged 10–14 years. It consists of 18 items covering pupil perceptions of liking, personal acceptance, inclusion, respect, encouragement for participation and response from peers and adults in school. Pupils respond to each item using a five-point scale (from 'Not at all true' to 'Completely true').

The *Belonging Scale* is an adapted version of the *PSSM Scale*, reduced from 18 to 12 items. It has been anglicised and simplified for use with children as young as 8 years of age, including those with special educational needs (Frederickson et al. 2007). Children are asked to rate statements such as 'I feel really happy at my school' on a three-point response scale: 'No not true', 'Not sure' and 'Yes true'.

Of particular importance to the inclusion of the *PSSM* in this booklet are recent findings linking scores on the measure with important aspects of child and adolescent mental health and psychological wellbeing, such as depressive symptoms in adolescence. Longitudinal research with twelve- to fourteen-year-olds in Australia found a predictive link from *PSSM* scores to future mental health problems (Shochet et al. 2006). For both boys and girls there was a predictive link with depressive symptoms on the *Child Depression Inventory* (Kovacs 2003). In addition, predictive links were found for girls with anxiety symptoms on the *Spence Children's Anxiety Scale* (Spence 1998) and for boys with the total difficulties score on the Goodman *Strengths and Difficulties Questionnaire* (Goodman 1997).

More specifically, links have been reported for vulnerable groups, such as those with refugee status or disabilities. Kia-Keating and Ellis (2007) found that school belonging, assessed by the *PSSM*, was positively related to self-efficacy and negatively related to depression in a sample of Somali adolescent refugees in the United States. McMahon et al. (2008) used the *PSSM* to investigate school belonging following the closure of a school for pupils with disabilities (aged 10–18 years) and transition to new inclusive school placements. High belonging scores were associated with academic self-efficacy and school satisfaction while low belonging scores were associated with depression.

Administration

The scales can be administered to individuals, small groups or whole-class groups. Where administered to whole-class groups, it is advisable for primary pupils in particular to have a second adult present who can circulate among the pupils, answer questions and provide additional help to those who need it.

It is good practice to inform pupils at the start of the session about the purpose of the survey and to emphasise that their honest opinion is sought, giving appropriate assurances.

For the *PSSM*, start both individual and group administration by reading out the instructions. Where pupils may have difficulty reading the questionnaire, the alternative wording in brackets should be used and the questionnaire read out to them:

> *Read each sentence and show (I will read each sentence and I would like you to show) how true it is for you by circling a number between 1 and 5. If you think it is* completely true *for you circle 5. If you think it is* not at all true *for you circle 1* (show the other options between 1–5 and read them out). *Are there any questions? Remember there are no right or wrong answers.*

For the *Belonging Scale* it is recommended that for group administration each item is read aloud while pupils follow the wording on their questionnaire sheet. The response categories can also be read out for the first few questions to help ensure that pupils understand what to do:

> *I will read each sentence and I would like you to think if it is true for you or not. If you think it is true you circle* YES, *if you think it is not true you circle* NO. *If you are not sure whether it is true or not you circle the* question mark. *Let's do the first one as a practice . . . I would like you tell me if this is true for you or not:* I feel happy drawing pictures.

Circulate round the class and check everyone has understood the task. Remind the children that there are no wrong or right answers and that they should put their hand up if there is anything they do not understand.

Scoring

To avoid the development of a 'response set' by the children, a number of items on each scale are phrased so that disagreement, and hence a low item score indicates a stronger sense of belonging (e.g. 'Sometimes I feel as if I don't belong here'). The scores for these items are reversed when calculating the child's score.

PSSM

1. In scoring the *PSSM*, reverse the scores on the negatively phrased items. The reverse scored items are 3, 6, 9, 12 and 16. For each of these items cross out the number circled by the pupil (unless it is 3, which does not need to be changed). Instead circle the reverse score as follows:
   - If the pupil circled 5, circle 1 instead.
   - If the pupil circled 4, circle 2 instead.
   - If the pupil circled 2, circle 4 instead.
   - If the pupil circled 1, circle 5 instead.
2. Add the scores for each item and divide by 18 to get the pupil's average score. This is their *PSSM* score.
3. The percentage of pupils in a class or school with a low belonging score can be calculated by dividing the number of pupils who score below '3' by the number of pupils who completed the measure and multiplying by 100.

Belonging

1. In scoring the *Belonging Scale*, allocate points to items 1, 2, 4, 6, 7, 11 and 12 as follows: 1 for 'No, not true', 2 for '? Not sure' and 3 for 'Yes, true'.
2. Reverse the scores on the negatively scored items (3, 5, 8, 9 and 10) allocating 3 for 'No, not true', 2 for '?, not sure' and 1 for 'Yes, true'.
3. Add the scores for each item and divide by 12 to get the pupil's average score. This is their *Belonging Scale* score.
4. The percentage of pupils in a class or school with a low belonging score can be calculated by dividing the number of pupils who score below '2' by the number of pupils who completed the measure and multiplying by 100.

Table 1.4 Percentage of pupils reporting 'low belonging' to school in a UK evaluation of inclusive practice

Sample	1 secondary school 202 pupils, Years–10	4 primary schools 526 pupils, Years 3–6
Percentage of pupils scoring below 3.0	21.3%	18.6%
Mean (SD)	3.50 (.67)	3.61 (.72)

Interpretation

For the *PSSM*, at an *individual/group* level, Goodenow (1993) advises that the scale midpoint of 3.0 may be regarded as a potential 'tipping point', below which a pupil is more negative than positive in their responses regarding the school. It is suggested that pupils scoring below this point may be regarded as potentially at risk in terms of their social inclusion or commitment to education.

At a school level the percentage of pupils falling below the scale mid-point can provide an indicator of the quality of school social relations as experienced by the pupils. This was the approach adopted in an OECD survey of 15-year-olds in 43 countries, using a six-item sense of belonging scale (rated on five points like the *PSSM*). Across most of the countries surveyed almost a quarter scored below the scale mid-point. A positive finding for the UK was that it had the lowest percentage of 'low belonging' (17 per cent). Data from an evaluation of inclusive educational practice in Buckinghamshire (Frederickson et al. 2004), shown in Table 1.4, indicates the relevance of using 25 per cent of pupils scoring below 3 on the *PSSM* as a criterion in triggering further investigation and intervention.

On the *Belonging Scale* the mid point (of 2 in this case) can be used to identify pupils who have a low sense of belonging and may be at risk of alienation. UK data from an inclusion study in Kent (Frederickson et al. 2007) is reported in Table 1.5. In this study 397 children from Year 3 to Year 6 classes in 14 schools took part. A mean score below 2 was obtained by 16 per cent of the sample.

Table 1.5 Number of schools in each percentage band on pupil 'low belonging' scores

Percentage of pupils scoring below 2.0	Number of primary schools (out of 14)
Less than 10%	2
10–24%	8
25% and more	4

Factors to consider in score interpretation:

Gender – differences are frequently reported, with girls reporting a stronger sense of belonging to their school than boys (Goodenow 1993; Hagborg 1994).

Social disadvantage – Goodenow (1993) found significant differences between a predominately middle-class suburban sample of pupils aged 10–14 years, where only 13 per cent reported a score at or below the scale midpoint of 3.0, and an economically disadvantaged urban sample where the percentage was more than three times higher, at 45.4 per cent.

Age is not usually found to make a difference to *PSSM* scores.

Studies that have compared the sense of belonging reported by pupils who have *special educational needs* and by their typically developing classmates

have nd significant differences (Hagborg 1998ckson et al. 2007). In both cases the authcented on the supportive nature of the inclusements.

Teal details

Good993) reported internal consistency reliabes of .77 to .88 for pupils aged 9–14 years *PSSM*. Hagborg (1994) reported modt–retest reliability of .78 over 4 weeks with e of 13-year-olds. Frederickson et al. (2007d high internal consistency reliability (.87) 2-item *Belonging Scale*.

A randity data is available. Goodenow (1993) founupils who joined a school during an acadr reported lower school membership than no had attended the school for longer. For fourteen-year-olds *PSSM* scores in the f of the academic year were found to correitively with end of year grades (.33) and effort ratings (.25), but negatively with abse8). Hagborg (1994) reported moderate correcross Year 6 to Year 13 of belonging with nce and attainment but moderate to sorrelations with all the Harter *Self-Percscales* (Harter 1985) including *Global Self* 5).

Featuthe school environment have also beeno have a significant association with schoging. In a study with pupils from Years 6 tonging was positively correlated with percof empathy and support from school staffatively correlated with interpersonal difficvith peers and adults in school (McNal. 2008). Among pupils in Years 5 and 6, sconging was positively associated with positgs of their classroom environment on the N Inventory[2] cohesion and satisfaction scaldahon et al. 2009). Scores on the difficpetitiveness and friction. Scales were negasociated with belonging.

Fredet al. (2007) found that scores on the *Belcale* for children in Years 3 to 6 were positively correlated with peer group acceptance, as assessed by the Social Inclusion Survey[3] and nomination on positive behavioural descriptors (such as 'cooperates') on the *Guess Who Peer Assessment* measure. Belonging scores were negatively correlated with peer rejection and scores on negative behavioural descriptors, such as Disrupts and Bullies.

Comparison with similar measures

O'Farrell and Morrison (2003) reported that in the motivation, risk and resilience literature a majority of the studies use the *PSSM* to assess school belonging. For pupils in Key Stage 4, the 6 belonging items used in the OECD Programme for International Student Assessment (Willms 2003) represent an alternative on which there is good national and international comparative data.

In contrast to approaches using self-rating of feelings or sense of belonging, Murdoch's (1999) behavioural measure of school alienation collects ratings from teachers on pupil engagement, attendance, class participation and work completion. Pupils self-rate the frequency with which four disciplinary sanctions have been applied to them: sent to the deputy head teacher, detention, in-school exclusion, fixed-term exclusion. While this approach will identify serious cases, urgently in need of intervention, schools are usually very well aware of these pupils. The advantage of the *PSSM* is that early signs of disconnection to school can be picked up.

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[2]The N Inventory' can be found in 'The Learning Environment' booklet of *Psychology in Education Portfolio* (GL Ant 1999).

[3]The Inclusion Survey' and the 'Guess Who Measure' can be found in the 'Social Skills and Emotional Intelligence' booklet of *Psychology in Education Portfolio* (GL Assessment 1999) available at www.gl-assessment.co.uk

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